



Note: 1. Submission only via E-CLASS is acceptable .

2. Do not plagiarize on doing your work, if we catch student doing so, he/she will get ZERO for the Assignment.

Lab Assignment -6 (Recursion)

1. Calculate the x^y using a tail recursive function.

2. Find sum of natural numbers using recursion.

(Hint: Natural numbers are a part of the number system which includes all the positive integers from 1 till infinity and are also used for counting purpose. It does not include zero (0). In fact, 1,2,3,4,5,6,7,8,9....., are also called counting numbers.)

3. Display Fibonacci Series Using Recursion.

4. Find G.C.D for two integers using recursion

(Hint: The greatest common divisor of 12 and 26 can be calculated as:

Divisors of 12 are 1, 2, 3, 4, 6, and 12.

Divisors of 26 are 1, 2, 13, and 26.

Common divisors of 12 and 26 are 1 and 2.

∴ The greatest common divisor of (12, 26) is 2.)

5. Write a program to compute $F(x,y)$ where,

$$F(x,y)=F(x-y,y)+1 \quad \text{if } y \leq x$$

$$\text{And } F(x,y) = 0 \quad \text{if } x < y$$